Journal of the Brazilian Society of Mechanical Sciences and Engineering (2026) 48:244
https://doi.org/10.1007/s40430-025-06221-5

TECHNICAL PAPER

Robust fractional-order synergetic control for quadcopter systems

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Received: 23 August 2025 / Accepted: 25 November 2025
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Abstract
Quadcopters are widely used in many fields due to their unique characteristics and simple structure. However, controlling a quadcopter remains challenging because it is an underactuated system. The main difficulties lie in achieving precise trajectory tracking and maintaining stability under disturbances and parameter uncertainties. This work investigates the development of a Fractional Order Synergetic Control (FOSC) approach optimized by the Equilibrium Optimizer (EO) algorithm to enhance the control performance of quadcopters under uncertain and disturbed conditions. The proposed method combines the advantages of fractional-order calculus with synergetic control theory, offering greater tuning flexibility, faster convergence, and reduced chattering. The controller parameters are automatically adjusted using the EO to improve optimal dynamic behavior. Closed-loop stability is verified through the Lyapunov direct method. To evaluate the effectiveness of the proposed FOSC technique, several simulations are carried out under different situations, including external disturbances, parametric uncertainties, and mass variation. A comparative study is also conducted to show the improvements of the proposed controller in tracking accuracy, response speed, and robustness. The obtained results confirm that the proposed FOSC provides superior trajectory tracking and disturbance rejection compared with conventional control approaches.

Keywords Quadcopters · Synergetic Control · Fractional order · Equilibrium optimizer · Uncertainties parameters · External disturbances

1 Introduction

Recent years have witnessed a surge of interest in Unmanned Aerial Vehicle (UAV) research [1]. These UAVs, also known as drones, provide significant flexibility for both military and civilian applications [2]. Technological advancements in areas such as electric power storage, embedded hardware design, material structures, electronics, computing sciences, and real-time processing have paved the way for the development of innovative drones [3]. Among these, the quadcopter stands out as one of the most promising UAV types, with a wide range of applications across various fields [4]. These applications include mailing and delivery, environmental protection, search and rescue, monitoring, reconnaissance within buildings, maintenance, agriculture, videography/photography, mining, and operations involving nuclear, biological, and chemical hazards [5]. Quadcopters have emerged as versatile UAVs capable of operating in various environments, including indoor and outdoor settings, as well as underwater, over water surfaces, on the ground, in the air, and even in space [6]. Unlike conventional helicopters, quadcopters offer several advantages that make them more appealing for a wide range of applications. Their lightweight structure and simple mechanical design enable stable hover flight, which is particularly beneficial in confined spaces [7].

Quadcopters can maneuver closely around targets, perform vertical take-offs, and navigate through cluttered, small areas with ease. This versatility allows them to assist

Technical Editor by Emilio C. N. Silva.

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Published online: 10 March 2026

humans in everyday tasks and handle a broad spectrum of activities, especially those that pose challenges and risks. However, despite their numerous benefits, quadcopters face some significant challenges related to propeller rotation and blade flapping, which can impact their performance and safety [8]. The control of quadcopters is inherently complex due to the strong coupling between position and attitude, which adds to the system's under-actuated nature. This complexity is further exacerbated by aerodynamic disturbances and sensitivity to wind gusts, which can lead to instability [9]. Addressing these challenges involves tackling three main issues:

(1) The dynamics of the quadcopter represent a coupled multi-input multi-output (MIMO) nonlinear system.
(2) The position and attitude dynamics are subject to unmodeled uncertainties, external perturbations, and parametric uncertainties.
(3) The presence of multiple-input time delays and time-varying states.

The nonlinearity of the system and the coupling dynamics make controlling a quadcopter a complicated and tough task. These factors add to the complexity and make this study topic both fascinating and challenging. Many control techniques have been created for the quadcopter system, involving classical methods such as: Linear Quadratic Regulator (LQR) [10–13], Proportional Integral Derivative (PID) controller [14, 15], backstepping control [16, 17], Sliding Mode Control (SMC) [18, 19], and intelligent control techniques such as: fuzzy logic control [20, 21], neural network control [22, 23], and fuzzy neural network control [24–26].

Sliding Mode Control is a nonlinear control approach based on adjusting the system dynamics to follow a predefined sliding surface. It ensures strong robustness regarding disturbances. However, its primary limitation is the chattering phenomenon caused by quick switching of the control signal within that surface. Reference [27] presents a dual-channel disturbance rejection control technique using SMC and an Extended State Observer (ESO) for quadrotor attitude regulation. This structure enhances robustness and improves disturbance estimation accuracy. However, the technique indicates a lack of fractional-order flexibility, potentially constraining tuning adaptability in complex situations. Researchers in [28] proposed an adaptive SMC strategy augmented with a Multi-Layer Perceptron (MLP) neural network for quadcopters. However, their approach still relies on integer-order dynamics, which limits tuning flexibility and can cause chattering under large disturbances. The application of fractional calculus in control engineering has lately surged, due to its capacity to provide more freedom and robustness relative to traditional integer-order methods. In [29], the authors investigated the implementation of two advanced controllers, namely the SMC and the FOSMC, applied to both integer- and fractional-order quadrotor systems. The study aimed to improve the control performance, robustness, and accuracy of the quadrotor by exploiting the benefits of fractional calculus in control design. The work presented in [30] integrates Fractional-Order Linear Active Disturbance Rejection Control (FOLADRC) with an enhanced Sparrow Search Algorithm (SSA) to form a robust attitude stabilization strategy for quadcopter UAV. This hybrid design leverages the disturbance rejection capabilities of LADRC and the tuning flexibility of fractional-order PID control. In reference [31], a novel Fractional-Order Global SMC (FOGSMC) method was introduced for improving trajectory tracking of quadcopter UAVs operating in uncertain and disturbed environments. By constructing a fractional-order sliding manifold, the approach ensures enhanced robustness and tracking precision. The authors of the work presented in [32] introduce a fractional-order controller integrated with ADRC to improve disturbance rejection and system responsiveness. In the work presented in [33], a Fixed-Time Fractional-Order SMC (FTFOSMC) strategy was developed to provide accurate trajectory tracking under external disturbances and system uncertainties.

Research on Synergetic Control (SC) for quadcopters is not widespread, making it a lot of potential development. SC avoids chattering, adapts to disturbances, and maintains stability in uncertain environments, making it ideal for modern, highly dynamic systems. In [34] a synergetic control strategy is designed for transformable quadcopters operating under external disturbances. In reference [35], a robust control framework was proposed for a morphing quadcopter with independently rotating arms and time-varying structural parameters. This control framework, based on synergetic theory, employs a Fast Terminal SC (FTSC) to address dynamic uncertainties, achieving fast convergence, improved tracking, and reduced chattering. In [36], a Fractional-Order Synergetic Control (FOSC) strategy was proposed for a 4-DOF robotic manipulator, integrating the flexibility of fractional calculus with the robustness of synergetic control. The FOSC ensures precise joint and workspace trajectory tracking, fast convergence, and chattering-free performance. However, the present study addresses this approach for quadrotor UAVs under disturbances and system uncertainties, a topic not previously explored. Despite several studies that have addressed quadcopter control using fractional-order or synergetic techniques independently, there remains a clear research gap in integrating both methods within a unified framework optimized by modern metaheuristic algorithms. The suggested FOSC, optimized by the EO, addresses this gap by integrating the tuning flexibility

of fractional calculus with the chattering-free and invariant characteristics of synergetic control. This hybrid structure enables faster convergence and enhanced robustness against parameter uncertainties and external disturbances, which distinguishes the present work from existing approaches in the literature.

The purpose of this work is to investigate the problem of trajectory tracking control for a quadcopter through the integration of SC and fractional calculus to enhance robustness and dynamic performance. Despite the remarkable progress achieved in quadcopter control design, the combination of SC and fractional calculus in UAV applications remains scarcely explored. Most existing approaches still rely on classical or integer-order control methods, which often fail to maintain robustness and stability under strong disturbances and parameter uncertainties. To overcome these limitations, this study proposes a FOSC scheme optimized using EO algorithm. The proposed controller merges the flexibility of fractional calculus with the robust and smooth characteristics of SC, leading to improved trajectory tracking accuracy, faster convergence, and enhanced robustness to system uncertainties and external disturbances. Furthermore, this study contributes new insights into the control of quadcopter UAVs by addressing aspects that were not considered in previous control design works.

The primary scientific contributions of this paper can be summarized as follows:

- The proposal of a FOSC strategy optimized by EO algorithm for quadrotor UAVs subjected to external disturbances and parametric uncertainties. This advanced control framework ensures robust and precise trajectory tracking while preserving smooth control actions and fast transient responses, while effectively removing the chattering phenomena associated with conventional control approaches.
- The integration of fractional calculus with synergetic control theory, providing additional tuning flexibility, enhanced adaptability, and improved robustness compared with traditional integer-order and standalone synergetic controllers.
- The controller parameters are tuned automatically using the EO optimization mechanism, which improves dynamic performance, accelerates convergence, and enhances disturbance rejection without the need for manual gain adjustment.
- A comprehensive stability verification is conducted using the Lyapunov direct method, confirming the global stability and robustness of the closed-loop system under external disturbances.
- A comparative performance assessment is performed against conventional control techniques, including SC and SMC. Simulation results clearly demonstrate the superior tracking accuracy, robustness, and convergence speed of the proposed FOSC controller.

The rest of this paper is organized as follows: Sect. 2 presents the quadcopter description and the dynamics model based on Newton–Euler equations, incorporating external disturbances, and reformulates it as an affine nonlinear system suitable for control design. Section 3 presents the developed control strategy, starting with the development of a synergetic controller and extending it to a fractional-order synergetic control framework. This section also includes the optimization of the controller parameters using the equilibrium optimizer algorithm, as well as the closed-loop stability analysis based on the Lyapunov direct method. Section 4 discusses the simulation results that validate the effectiveness of the proposed controller under various scenarios involving uncertainties and disturbances. Finally, Sect. 5 concludes the paper.

2 Quadcopters dynamics modelling

The quadrotor features four propellers arranged in a cross configuration, with two pairs rotating in opposing directions. Altering rotor speeds enables the modification of lift forces and the generation of motion. Consequently, uniformly augmenting or diminishing the velocities of the four propellers produces vertical movement. Adjusting the speeds of the 2 and 4 propellers inversely results in roll rotation accompanied by lateral movement. Pitch rotation and the associated lateral movement arise from the inverse modification of the speeds of propellers 1 and 3. Yaw rotation is more nuanced, as it arises from the disparity in counter-torque between each set of propellers. The quadcopter configuration is shown in Fig. 1.

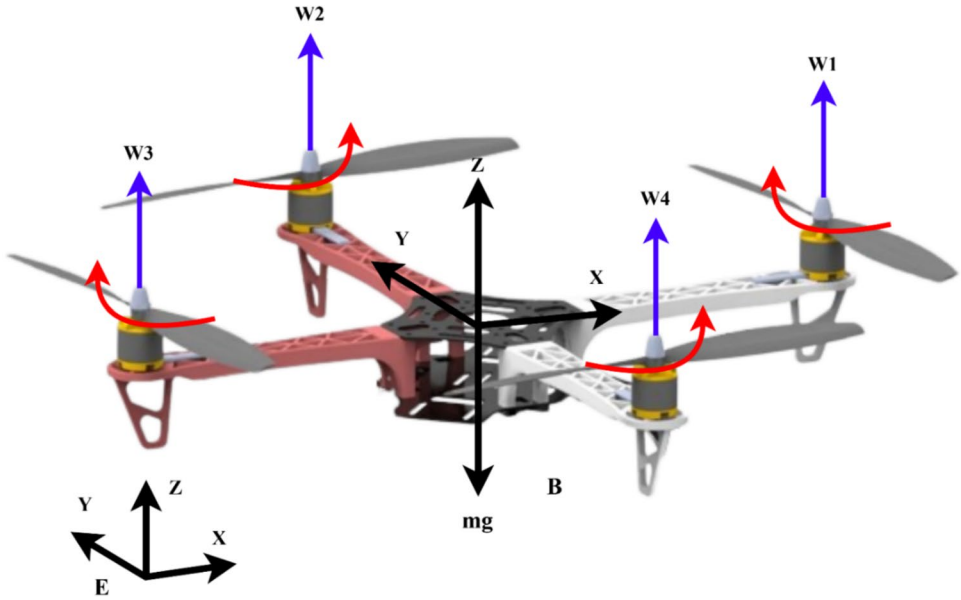
To simplify the dynamics modelling, the following assumptions are included:

1. The quadrotor configuration is rigid and symmetrical.
2. The propellers are rigid
3. Thrust and drag forces are related to the square of the rotational speed of the propellers.

Given these assumptions, the flight dynamics can be characterized as those of a rigid body subjected to aerodynamic forces and moments resulting from the propellers' rotation. Employing the Newton–Euler formalism, the equations of dynamics are articulated as follows [37]:

$$\begin{cases} m\ddot{\zeta} = F_f + F_d + F_g \\ J\dot{\Omega} = -\Omega \wedge J\Omega + \Gamma_f - \Gamma_a - \Gamma_g \end{cases} \tag{1}$$

Fig. 1 Quadcopter configuration



where ζ denotes the position of the quadrotor's center of mass with respect to the inertial frame E, m represents the total mass of the vehicle, and $J \in \mathbb{R}^{3\times 3}$ is the symmetric positive-definite inertia matrix of the quadrotor defined relative to the body-fixed frame:

$$J = \begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix} \tag{2}$$

$\mathbf{\Omega}$ represents the angular velocity vector of the quadrotor expressed in the body-fixed reference frame, defined as:

$$\Omega = \begin{pmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \cos\theta\sin\phi \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{pmatrix} \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} \tag{3}$$

where ϕ, θ, and ψ denote the roll, pitch, and yaw angles of the quadrotor, respectively.

In the case where the quadrotor undergoes small-amplitude angular motions, the angular velocity $\mathbf{\Omega}$ can be approximated as $\mathbf{\Omega} \approx [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$. The term $\mathbf{F_f}$ denotes the resultant force generated by the four rotors, defined as:

$$F_f = \begin{pmatrix} \cos\phi\cos\psi\sin\theta + \sin\phi\sin\psi \\ \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ \cos\phi\cos\theta \end{pmatrix} \sum_{\mathrm{i}=1}^{4} F_i \tag{4}$$

$$F_i = K_p \omega_i^2 \tag{5}$$

where K_p is the lift coefficient, and ω_i represents the angular speed of the ith rotor. The term $\mathbf{F_d}$ denotes the resultant drag force components along the (X, Y, Z) axes, defined as:

$$F_d = \begin{pmatrix} -K_{fdx} & 0 & 0 \\ 0 & -K_{fdy} & 0 \\ 0 & 0 & -K_{fdz} \end{pmatrix} \dot{\zeta} \tag{6}$$

where K_{fdx}, K_{fdy}, and K_{fdz} are the positive translational drag coefficients. The gravitational force $\mathbf{F_g}$ is expressed as follows:

$$F_g = \begin{pmatrix} 0 \\ 0 \\ -mg \end{pmatrix} \tag{7}$$

$\Gamma_\mathbf{f}$ represents the total moment generated by the quadrotor with respect to the body-fixed reference frame, defined as:

$$\Gamma_f = \begin{pmatrix} l\,(F_3 - F_1) \\ l\,(F_4 - F_2) \\ C_D\left(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2\right) \end{pmatrix} \tag{8}$$

where l denotes the distance between the quadrotor's center of mass and the rotation axis of each propeller, and C_D is the drag coefficient. The term $\Gamma_\mathbf{a}$ represents the resultant aerodynamic friction torques, defined as:

$$\Gamma_a = \begin{pmatrix} K_{fax} & 0 & 0 \\ 0 & K_{fay} & 0 \\ 0 & 0 & K_{faz} \end{pmatrix} \|\Omega\|^2 \tag{9}$$

where K_{fax}, K_{fay}, and K_{faz} are the aerodynamic friction coefficients. The term $\Gamma_\mathbf{g}$ denotes the resultant torques generated by the gyroscopic effects:

$$\Gamma_g = \sum_{i=1}^{4} \Omega \wedge J_r \begin{pmatrix} 0 \\ 0 \\ (-1)^{i+1}\omega_i \end{pmatrix} \tag{10}$$

where J_r denotes the rotor moment of inertia.

In realistic flight conditions, quadcopters are inevitably affected by modeling uncertainties and external disturbances such as aerodynamic perturbations or wind gusts. These disturbances represent unmodeled dynamics and environmental forces acting on the system.

Consequently, the complete dynamic model governing the quadcopter can be expressed as follows:

$$\begin{cases} \ddot{\phi} = \frac{1}{I_x}\left(\dot{\theta}\dot{\psi}(I_y - I_z) - J_r\dot{\theta}\Omega_r - K_{fax}\dot{\phi}^2 + lU_2\right) + d_\phi \\ \ddot{\theta} = \frac{1}{I_y}\left(\dot{\phi}\dot{\psi}(I_z - I_x) - J_r\dot{\phi}\Omega_r - K_{fay}\dot{\theta}^2 + lU_3\right) + d_\theta \\ \ddot{\psi} = \frac{1}{I_z}\left(\dot{\phi}\dot{\theta}\left(I_x - I_y\right) - K_{faz}\dot{\psi}^2 + U_4\right) + d_\psi \\ \ddot{x} = \frac{1}{m}\left(U_xU_1 - K_{fdx}\dot{x}\right) + d_x \\ \ddot{y} = \frac{1}{m}\left(U_yU_1 - K_{fdy}\dot{y}\right) + d_y \\ \ddot{z} = \frac{1}{m}\left((cos(\theta)cos(\phi))\,U_1 - K_{fdz}\dot{z}\right) - g + d_z \end{cases} \tag{11}$$

with:

$\Omega_r = w_1 - w_2 + w_3 - w_4$ and U_x, U_y are the virtual control inputs defined as:

$$\begin{cases} U_x = c\phi s\theta c\psi + s\phi s\psi \\ U_y = c\phi s\theta s\psi - s\phi c\psi \end{cases} \tag{12}$$

where $c\phi \equiv \cos\phi$ and $s\phi \equiv sin\phi$ (same for $c\theta, s\theta, c\psi, s\psi$), d_ϕ, d_θ, d_ψ, d_x, d_y and d_z represent the bounded disturbances. U_1 represents the total lift force on the body in the z-axis; U_2, U_3 and U_4 are the roll, pitch and yaw torque, respectively, where:

$$\begin{pmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{pmatrix} = \begin{pmatrix} K_p & K_p & K_p & K_p \\ -K_p & 0 & K_p & 0 \\ 0 & -K_p & 0 & K_p \\ C_D & -C_D & C_D & -C_D \end{pmatrix} \begin{pmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{pmatrix} \tag{13}$$

From the Eq. (12) we can extract the expressions of the nonholonomic constraints which represent the relations between the two angles ϕ and θ with the components of the acceleration.

$$\begin{cases} \phi = \arcsin(U_x s\psi - U_y c\psi) \\ \theta = \arcsin((U_x c\psi + U_y s\psi)/c\phi) \end{cases} \tag{14}$$

Consequently, Eq. (14) can be used to formulate the nonholonomic constraints of the desired angles ϕ_d and θ_d as follows

$$\begin{cases} \phi_d = \arcsin(U_x s\psi_d - U_y c\psi_d) \\ \theta_d = \arcsin((U_x c\psi_d + U_y s\psi_d)/c\phi_d) \end{cases} \tag{15}$$

The model given by (11) represents the standard rigid-body dynamics of a quadcopter. This formulation is widely adopted in recent UAV research and has been validated in numerous studies focused on nonlinear and robust control. It accurately captures the essential translational and rotational coupling effects while maintaining a level of simplicity suitable for controller design and numerical simulation. Therefore, the adopted model can be regarded as a verified reference for simulation-based analysis and is thus suitable for accurate validation of the proposed control strategy, as reported in several recent works [16, 28, 38–40]. This model can be generalized to a class of affine systems Σ given below:

$$\Sigma : \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x) + g(x)U + d \end{cases} \tag{16}$$

where $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \in \mathbb{R}^{6\times 2}$ is the state vector, $x_1 = [\phi \quad \theta \quad \psi \quad x \quad y \quad z]^T \in \mathbb{R}^6$, $x_2 = \begin{bmatrix} \dot{\phi} & \dot{\theta} & \dot{\psi} & \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^T \in \mathbb{R}^6$,$U = \begin{bmatrix} U_2 & U_3 & U_4 & U_x & U_y & U_1 \end{bmatrix}^T$ is the control input vector and $d = \begin{bmatrix} d_\phi & d_\theta & d_\psi & d_x & d_y & d_z \end{bmatrix} \in \mathbb{R}^6$ represents the bounded disturbances, $f(x)$ and $g(x)$ are nonlinear functions describing the system, where:

$$f(x) = \begin{pmatrix} \frac{1}{I_x}\left(\dot{\theta}\dot{\psi}(I_y - I_z) - J_r\dot{\theta}\Omega_r - K_{fax}\dot{\phi}^2\right) \\ \frac{1}{I_x}\left(\dot{\phi}\dot{\psi}(I_z - I_x) - J_r\dot{\phi}\Omega_r - K_{fay}\dot{\theta}^2\right) \\ \frac{1}{I_z}\left(\dot{\phi}\dot{\theta}\left(I_x - I_y\right) - K_{faz}\dot{\psi}^2\right) \\ \frac{1}{m}\left(-K_{fdx}\dot{x}\right) \\ \frac{1}{m}\left(-K_{fdy}\dot{y}\right) \\ \frac{1}{m}\left(-K_{fdz}\dot{z}\right) - g \end{pmatrix}, \; and$$

$$g(x) = \begin{pmatrix} \frac{l}{I_x} \\ \frac{l}{I_y} \\ \frac{1}{I_z} \\ \frac{U_1}{m} \\ \frac{U_1}{m} \\ \frac{\cos(\theta)\cos(\phi)}{m} \end{pmatrix}.$$

3 Controllers design

This paper aims to design a robust nonlinear controller based on synergetic control theory to ensure precise trajectory tracking of the quadcopter despite the presence of external disturbances and parametric uncertainties, such as variations in mass and inertia resulting from the vehicle's structural complexity.

3.1 Control architecture

The quadcopter system includes 4 direct control inputs, U_1, U_2, U_3 and U_4, in addition to 2 virtual controllers, U_x and U_y, with 6 degrees of freedom. The system is divided into 2 cascade subsystems: one for the position control, which corresponds to the outer loop, and the second for the attitude control, which corresponds to the inner loop. By separating the position and attitude dynamics, the quadcopter is split into two coupled subsystems. The rotation dynamics are not impacted by the translation components; however, the translational dynamics are impacted by angles. The signal U_1 controls the altitude, while the two additional translation variables, x and y, are controlled indirectly by controlling the 2 rotation angles (φ and θ) using the virtual controllers U_x and U_y. The control structure is shown in Fig. 2.

3.2 Synergetic control

Synergetic control is a nonlinear control approach based on the invariance principle. It operates in a similar way to sliding mode control but without the chattering effect. This technique has been successfully applied to a wide range of nonlinear and high-order dynamic systems.

Definition 1: The control vector U, drives the system trajectories from any initial condition toward an invariant manifold and then to the equilibrium point. The control law is expressed as a function of macro-variables ψ, which depend on the system states. These variables are designed so that the following condition holds [41]:

$$\mathrm{K}\dot{\psi} + S(\psi) = 0 \tag{17}$$

where K is a design variable that determines the convergence rate of the macro variable ψ to the invariant manifold $\psi = 0$, and $S(\psi)$ is a smooth differentiable function of ψ that is selected in such a manner that:

$$S(0) = 0$$

Invertible and differentiable;

$$S(\psi)\,\psi > 0, \forall \psi \neq 0$$

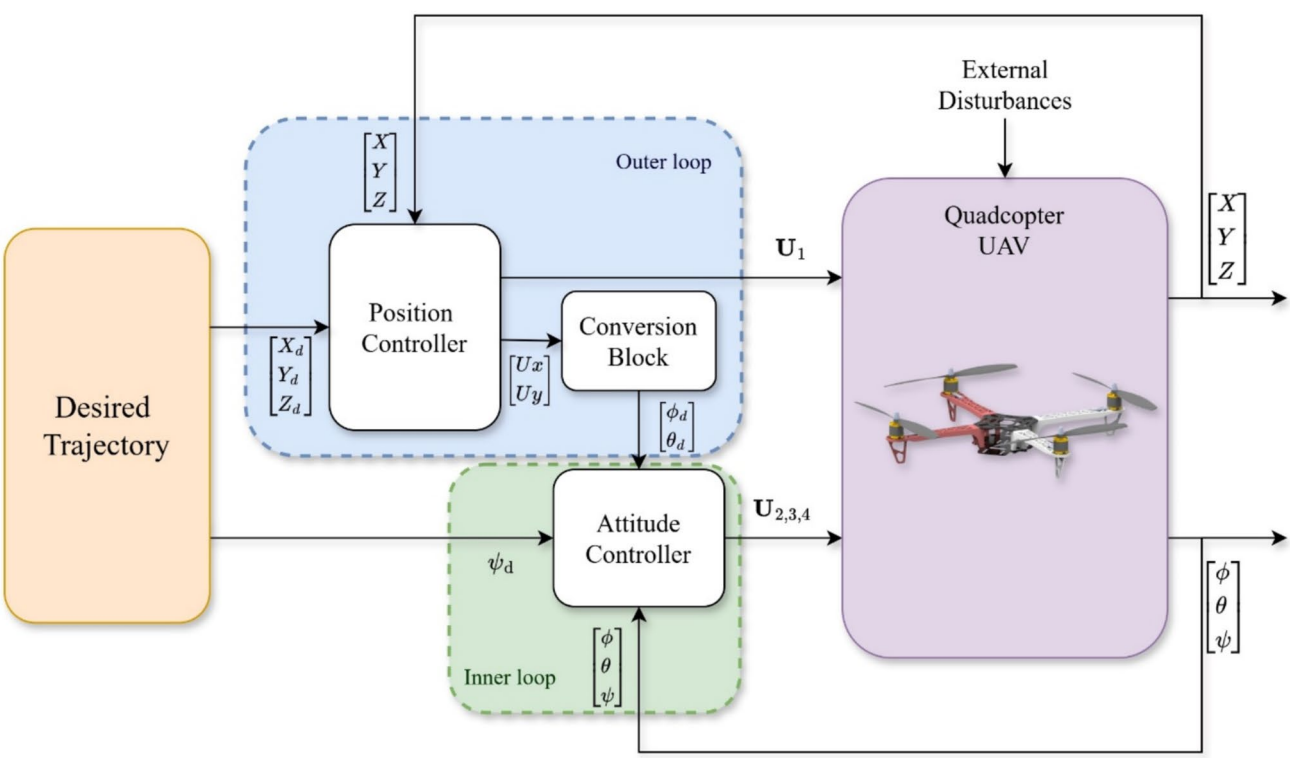


Fig. 2 Control structure

Selecting the macro variable function ψ as a function of the tracking error:

$$\psi = \Gamma e + \dot{e} \tag{18}$$

where $\Gamma = \text{diag}\left[\begin{array}{cccccc} \Gamma_\phi & \Gamma_\theta & \Gamma_\psi & \Gamma_x & \Gamma_y & \Gamma_z \end{array}\right] \in \mathbb{R}_{+*}^{6\times 6}$.

The above conditions can be achieved by:

$$S(\psi) = \psi \tag{19}$$

Proof: We demonstrate that our selected macro-variable function respects the previously outlined conditions.

- $S(0) = 0$, is clear according to (18) and (19).
- $S(\psi)$ satisfies invertibility, where $\dot{S}(\psi) > 0$ and $S(\psi)$ is a monotone function.
- By (6), we have $S(\psi)\psi = \psi^T\psi > 0, \forall\psi \neq 0$.

For this macro variable, we generally select the dynamic evolution function as follows:

$$\mathrm{K}\dot{\psi} + \psi = 0 \tag{20}$$

Where $\mathrm{K} = \text{diag}\left[\begin{array}{cccccc} \mathrm{K}_\phi & \mathrm{K}_\theta & \mathrm{K}_\psi & \mathrm{K}_x & \mathrm{K}_y & \mathrm{K}_z \end{array}\right] \in \mathbb{R}_{+*}^{6\times 6}$. It is characterized by the ability to implement velocity convergence defined by the designer at the targeted equilibrium point.

Let:

$$\begin{cases} e = x_{1d} - x_1 \\ \dot{e} = \dot{x}_{1d} - x_2 \end{cases} \tag{21}$$

where $x_{1d} = [\phi_d \quad \theta_d \quad \psi_d \quad x_d \quad y_d \quad z_d]^T \in \mathbb{R}^6$ is the desired states vector.

The time derivative of (18) given as:

$$\dot{\psi} = \Gamma\left(\dot{x}_{1d} - x_2\right) + \left(\ddot{x}_{1d} - \dot{x}_2\right) \tag{22}$$

From (20) and (22), we obtain:

$$\Gamma(\dot{x}_{1d} - x_2) + \ddot{x}_{1d} - f(x) - g(x)U = -\mathrm{K}^{-1}\psi \tag{23}$$

Then the control law is given as:

$$\begin{aligned} U = g(x)^{-1}[&\mathrm{K}^{-1}(\Gamma(x_{1d} - x_1) \\ &+ (\dot{x}_{1d} - x_2)) + \Gamma(\dot{x}_{1d} - x_2) - f(x) + \ddot{x}_{1d}] \end{aligned} \tag{24}$$

Stability analysis: This analysis includes evaluating the stability of SC by examining a Lyapunov function candidate, defined as:

$$V = \frac{1}{2}\psi^T\psi \tag{25}$$

Then:

$$\dot{V} = \psi^T\dot{\psi} \tag{26}$$

Which implies

$$\dot{V} = \psi^T\left(\Gamma\left(\dot{x}_{1d} - x_2\right) + \left(\ddot{x}_{1d} - \dot{x}_2\right)\right) \tag{27}$$

Then:

$$\dot{V} = \psi^T(\Gamma(\dot{x}_{1d} - x_2) + \ddot{x}_{1d} - f(x) - g(x)U - d) \tag{28}$$

Replacing the control law U in (28) by its expression (24), we find

$$\dot{V} = \psi^T(-\mathrm{K}^{-1}\psi - d) \tag{29}$$

Condition 1: We assume that d is bounded, such that

$$|d| < \zeta\,|\psi| \tag{30}$$

Which implies

$$\dot{V} \leq \psi^T(-\mathrm{K}^{-1}\psi + \zeta\psi) \tag{31}$$

$$\dot{V} \leq -\psi^T(\mathrm{K}^{-1} - \zeta)\psi \tag{32}$$

Condition 2: We assume that

$$\mathrm{K}^{-1} > \zeta \tag{33}$$

Hence, stability is ensured by the negative derivative of the Lyapunov function for $\psi \neq 0$ and $\mathrm{K}^{-1} > \zeta$.

To improve the performance of the synergetic controller and achieve finite-time state convergence, a fractional synergetic controller is proposed in the following.

3.3 Fractional-order synergetic control

Fractional calculus extends the concept of conventional integer-order calculus by introducing derivatives and integrals of non-integer (real or complex) order. Originating from the early studies of Leibniz and L'Hôpital, this mathematical framework provides a flexible means to model complex dynamic systems. Its capability to represent memory and hereditary effects makes it an effective tool in various fields, including physics, economics, and control engineering [42, 43]. Several formulations of fractional calculus exist in the literature, such as the Grunwald–Letnikov (GL),

Riemann–Liouville (RL), and Caputo definitions. Among these, the Riemann–Liouville (RL) definition is the most commonly employed because of its generality and suitability for control applications. It is defined as follows:

The α th-order Riemann–Liouville fractional derivative is:

$$\begin{aligned} {}_{t_0}D_t^{\alpha} f(t) &= \frac{\mathrm{d}^{\alpha} f(t)}{\mathrm{d}t^{\alpha}} \\ &= \frac{1}{\Gamma(m-\alpha)} \frac{\mathrm{d}^m}{\mathrm{d}t^m} \int_{t_0}^{t} \frac{f(\tau)}{(t-\tau)^{\alpha-m+1}} \mathrm{d}\tau \end{aligned} \tag{34}$$

where m is considered as any integer, $\alpha \in \mathbb{R}^+$ is the fractional order of the derivative, where $m-1 < \alpha \leq m$, Γ is Euler's Gamma function and t_0 is the initial time. The Euler's Gamma function is defined as:

$$\Gamma(\alpha) = \int_0^{\infty} \mathrm{e}^{-t} t^{\alpha-1} \mathrm{d}t \tag{35}$$

The definition of the fractional order integral of Riemann–Liouville is given by:

$${}_{t_0}I_t^{\alpha} f(t) = \frac{1}{\Gamma(\alpha)} \int_{t_0}^{t} \frac{f(\tau)}{(t-\tau)^{1-\alpha}} \mathrm{d}\tau \tag{36}$$

The FOSC combines the advantages of fractional calculus and SC theory to improve the performance of quadrotor systems. The fractional-order term provides additional tuning flexibility, allowing more accurate adjustment of the controller parameters and better adaptability to dynamic variations. The SC ensures smooth convergence of system trajectories and effectively avoids the chattering phenomenon commonly observed in SMC. This combination improves robustness against external disturbances and parameter uncertainties while maintaining fast response and accurate trajectory tracking.

Let select the fractional macro-variable function as:

$$\psi = \dot{e} + \Gamma_1 e + \Gamma_2 I^{\alpha} e \tag{37}$$

where $\Gamma_1 = \mathrm{diag}\left[\begin{array}{cccccc} \Gamma_{1\phi} & \Gamma_{1\theta} & \Gamma_{1\psi} & \Gamma_{1x} & \Gamma_{1y} & \Gamma_{1z} \end{array}\right] \in \mathbb{R}_{+*}^{6\times 6}$ and $\Gamma_2 = \mathrm{diag}\left[\begin{array}{cccccc} \Gamma_{2\phi} & \Gamma_{2\theta} & \Gamma_{2\psi} & \Gamma_{2x} & \Gamma_{2y} & \Gamma_{2z} \end{array}\right] \in \mathbb{R}_{+*}^{6\times 6}$.

Condition 1 and **Condition 2** are satisfied by selection $S(\psi)$ as follows:

$$S(\psi) = \psi \tag{38}$$

Proof: We prove here that our chosen macro-variable function satisfies the following conditions.

- $S(0) = 0$ Is clear according to (37) and(38).
- $S(\psi)$ Satisfies invertibility, where $\dot{S}(\psi) > 0$ and $S(\psi)$ is a monotone function.
- By (38), we have $S(\psi)\psi = \psi^T \psi > 0$, $\forall \psi \neq 0$.

For this fractional macro-variable, we typically chose the dynamic evolution function as:

$$\mathrm{K}\dot{\psi} + \psi = 0 \tag{39}$$

where $\mathrm{K} > 0$.

Let:

$$\begin{cases} e = x_{1d} - x_1 \\ \dot{e} = \dot{x}_{1d} - x_2 \end{cases} \tag{40}$$

The time derivative of (37) is given as:

$$\dot{\psi} = \ddot{e} + \Gamma_1 \dot{e} + \Gamma_2 D^{1-\alpha} e \tag{41}$$

$$\begin{aligned} \dot{\psi} = &(\ddot{x}_{1d} - \dot{x}_2) + \Gamma_1 (\dot{x}_{1d} - x_2) \\ &+ \Gamma_2 D^{1-\alpha} (x_{1d} - x_1) \end{aligned} \tag{42}$$

From (39) and (42), we obtain:

$$\begin{aligned} &\ddot{x}_{1d} - f(x) - g(x)U + \Gamma_1 (\dot{x}_{1d} - x_2) \\ &\quad + \Gamma_2 D^{1-\alpha} (x_{1d} - x_1) = -\mathrm{K}^{-1}\psi \end{aligned} \tag{43}$$

Then the control law is given as:

$$\begin{aligned} U = &g(x)^{-1}[\mathrm{K}^{-1}\psi + \Gamma_1(\dot{x}_{1d} - x_2) \\ &+ \Gamma_2 D^{1-\alpha}(x_{1d} - x_1) - f(x) + \ddot{x}_{1d}] \end{aligned} \tag{44}$$

Stability analysis: This analysis includes evaluating the stability of our controller by examining a Lyapunov function candidate, defined as:

$$V = \frac{1}{2}\psi^T \psi \tag{45}$$

Then:

$$\dot{V} = \psi^T \dot{\psi} \tag{46}$$

Which implies

$$\dot{V} = \psi^T \begin{pmatrix} (\ddot{x}_{1d} - \dot{x}_2) + \Gamma_1 (\dot{x}_{1d} - x_2) \\ + \Gamma_2 D^{1-\alpha} (x_{1d} - x_1) \end{pmatrix} \tag{47}$$

Then:

$$\dot{V} = \psi^T \begin{pmatrix} (\ddot{x}_{1d} - f(x) - g(x)U - d) \\ +\Gamma_1 (\dot{x}_{1d} - x_2) + \Gamma_2 D^{1-\alpha} (x_{1d} - x_1) \end{pmatrix} \quad (48)$$

Replacing the control law U in (48) by its expression (44), we find

$$\dot{V} = \psi^T(-\mathrm{K}^{-1}\psi - d) \quad (49)$$

Based on **Condition 1**, we find

$$\dot{V} \leq \psi^T(-\mathrm{K}^{-1}\psi + \zeta\psi) \quad (50)$$

$$\dot{V} \leq -\psi^T(\mathrm{K}^{-1} - \zeta)\psi \quad (51)$$

By **Condition 2,** the stability is ensured by the negative derivative of the Lyapunov function for $\psi \neq 0$ and $\mathrm{K}^{-1} > \zeta$.

3.4 FOSC tuned by EO

Metaheuristic algorithms are powerful tools meant to solve difficult optimization problems, especially when traditional methods fall short due to excessive dimensionality or non-linearity. They provide approximate answers with modest computational effort and are widely employed in different disciplines, from engineering to artificial intelligence. In the work [44], the authors employed a genetic algorithm (GA) to optimize the weights of a Robust LQR Servomechanism (RSLQR) for UAV flight control systems. The authors in [45] employed a self-tuning particle swarm optimization (PSO) fuzzy PID controller, where PSO optimizes the quantization and scaling factors of the fuzzy PID algorithm, enhancing the robustness and accuracy of manipulator positioning against disturbances and parameter variations. In [46], the authors employed an improved grey wolf optimizer with weighting functions (IGWO-WFs) specifically applied to UAVs path planning.

To find the optimal FOSC parameters, we will apply a metaheuristic approach based on equilibrium optimizer (EO) [47]. This approach efficiently checks the search space, balancing exploration and exploitation to uncover high-performance control settings. By employing equilibrium-based techniques systems [48].

The EO algorithm optimizes the parameters of the FOSC, as specified in (44). The following are the typically cited equations utilized in the EO framework. The concentration of EO particles is revised according to the following equation:

$$\mathbf{C} = \mathbf{C}_{eq} + (\mathbf{C}_0 - \mathbf{C}_{eq}) \circ \mathbf{F} + (\frac{\mathbf{G}}{\lambda \circ V}) \circ (\mathbf{1} - \mathbf{F}) \quad (52)$$

Table 1 Physical parameters of the quadrotor

Parameters	Values	Parameters	Values
m	0.486 (Kg)	K_{fdx}, K_{fdy}, K_{fdz}	5.567×10^{-4} (N/m/s), 5.567×10^{-4} (N/m/s), 6.354×10^{-4} (N/m/s)
I_x, I_y, I_z	3.8278 × 10 − 3 (N.m/rad/s^2), 3.8278 × 10 − 3 (N.m/rad/s^2), 7.6566 × 10 − 3(N.m/rad/s^2)	K_{fax}, K_{fay}, K_{faz}	5.567×10^{-4} (N/rad/s), 5.567×10^{-4} (N/rad/s), 6.354×10^{-4} (N/rad/s)
J_r	2.8385×10^{-5} (Kg.m^2)	l,g	0.25 (m),9.81(m/s^2)
Kp	2.9842 × 10 − 5 (N.m/rad/s)	Cd	3.2320 × 10 − 7 (N.m/rad/s)

Table 2 Parameters of the EO algorithm

Parameters	Values	Parameters	Values
p	19	max_iter	500
ϵ	0	the number of independent runs	1
Number of particles	30	a_1	2
a_2	1	$\mathbf{G}_p$	0.5

$\mathbf{C} \in \mathbb{R}^p$ denotes the updated concentration, $\mathbf{C}_{eq} \in \mathbb{R}^p$ signifies the equilibrium concentration, $\mathbf{C}_0 \in \mathbb{R}^p$ indicates the current concentration, $\mathbf{F} \in \mathbb{R}^p$ represents the exponential term, $\mathbf{G} \in \mathbb{R}^p$ is the generation rate vector, $\lambda \in \mathbb{R}^p$ is a turnover rate vector with components between 0 and 1, p denotes the number of parameters to be optimized, $V \in \mathbb{R}$ and is the control volume, $\circ$ signifies the product.

The exponential term $\mathbf{F}$ is employed to equilibrate exploration and exploitation throughout the optimization method:

$$\mathbf{F} = a_1 sign(\mathbf{r} - 0.5)[e^{-\lambda \bar{t}} - 1] \quad (53)$$

with:

$$\bar{t} = \left(1 - \frac{\mathrm{Iter}}{max_iter}\right)^{(a_2 - \frac{\mathrm{Iter}}{\mathrm{max_iter}})} \quad (54)$$

where $\{a_1, a_2\} \in \mathbb{R}$ are constant values, max_iter specifies the maximum number of iterations, $\bar{t}$ indicates the time that varies as a function of each iteration ($Iter$), and $\mathbf{r} \in \mathbb{R}^p$ is a random vector with components ranging from 0 to 1. This term brings randomization into the algorithm, allowing for exploration of the search space.

The generation rate term is used as follows to improve the algorithm's capacity for exploration and exploitation:

$$\mathbf{G} = \mathbf{G}_0 \circ \mathbf{F} \quad (55)$$

with

$$\mathbf{G}_0 = \mathbf{G}_{CP} \circ (\mathbf{C}_{eq} - \lambda \circ \mathbf{C}) \tag{56}$$

such that

$$\mathbf{G}_{CP} = \begin{cases} 0.5r_1 & , r_2 \geq \mathbf{G}_p \\ \mathbf{0} & , r_2 < \mathbf{G}_p \end{cases} \tag{57}$$

where $\mathbf{G}_0 \in \mathbb{R}^p$ is the initial generation rate vector, $\mathbf{G}_{CP} \in \mathbb{R}^p$ represents the parameter that regulates the generation rate, $\mathbf{G}_p \in \mathbb{R}$ is the generation probability, and $\{r_1, r_2\} \in \mathbb{R}$ are random values between 0 and 1, The following equation delineates the objective function applied in the optimization of controller parameters, expressed as:

$$\begin{aligned} J = &\int_0^{t_f} \sqrt{(\phi_d - \phi)^2} dt + \int_0^{t_f} \sqrt{(\theta_d - \theta)^2} dt + \int_0^{t_f} \sqrt{(\psi_d - \psi)^2} dt \\ &+ \int_0^{t_f} \sqrt{(x_d - x)^2} dt + \int_0^{t_f} \sqrt{(y_d - y)^2} dt + \int_0^{t_f} \sqrt{(z_d - z)^2} dt \end{aligned} \tag{58}$$

The sequential procedure of the EO algorithm unfolds over the following steps [49]:

Step 1: initialization

- Select a desired trajectory.
- Determine the precision value ϵ, the number of independent runs, number of particles, and the maximum iteration number max_iter.
- Set the number of FOSC parameters p.
- Initialize the parameters in a random manner.

Step 2: cost function evaluation and concentration updating equation

- Compute the objective function using (58).
- Construct the equilibrium pool vector from the equilibrium concentration.

Step 3: Accomplish memory saving (if $Iter > 1$).

- Set time $\bar{t}$ according to (54).
- Select a single candidate randomly from the equilibrium pool vector.
- Generate random vectors for λ and $\mathbf{r}$.
- Construct $\mathbf{F}$, $\mathbf{G}_{CP}$ and $\mathbf{G}$ using (53), (57) and (55).

Step4: termination criteria

- If the cost function drops below a specified minimal tolerance value ϵ, or the max_iter is reached:
  - Select the optimal solution, which is the equilibrium pool, and complete the algorithm.

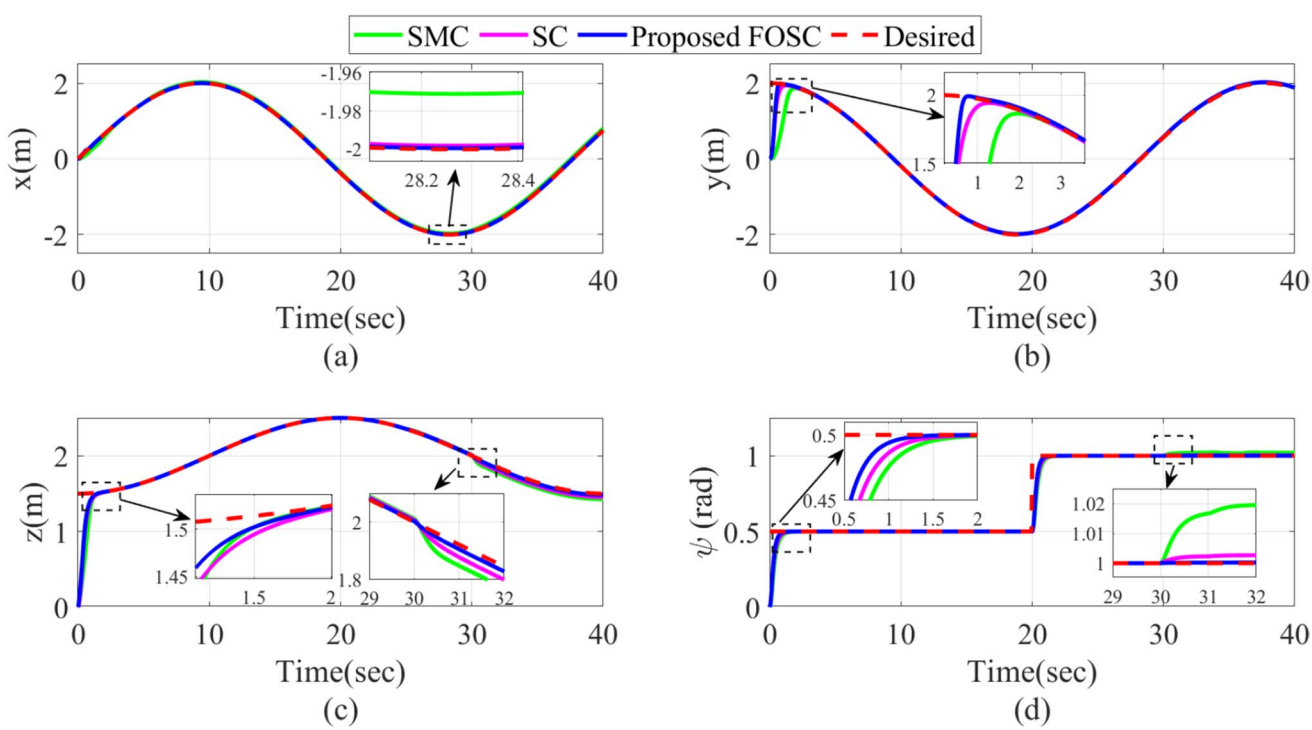


Fig. 3 Time responses of the quadcopter positions (*x, y, z*) and yaw angle (ψ) (scenario 1)

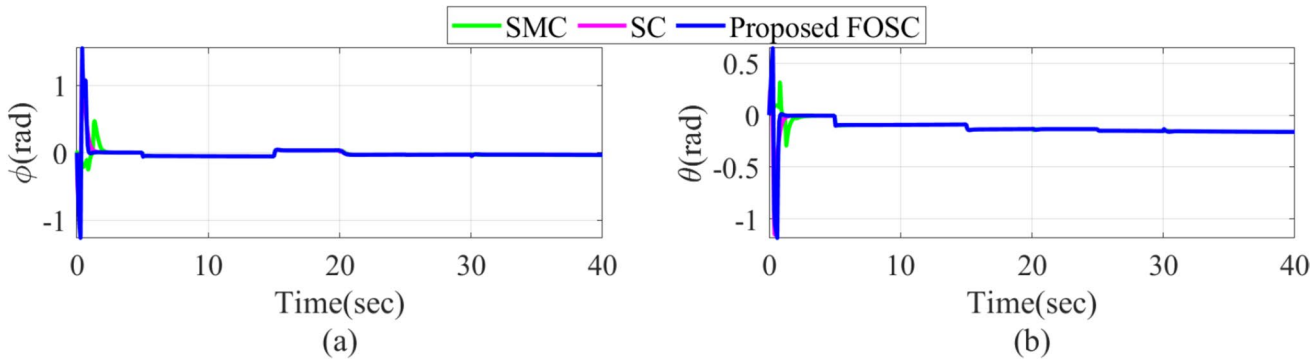


Fig. 4 Roll (φ) and pitch (θ) angular responses (scenario 1)

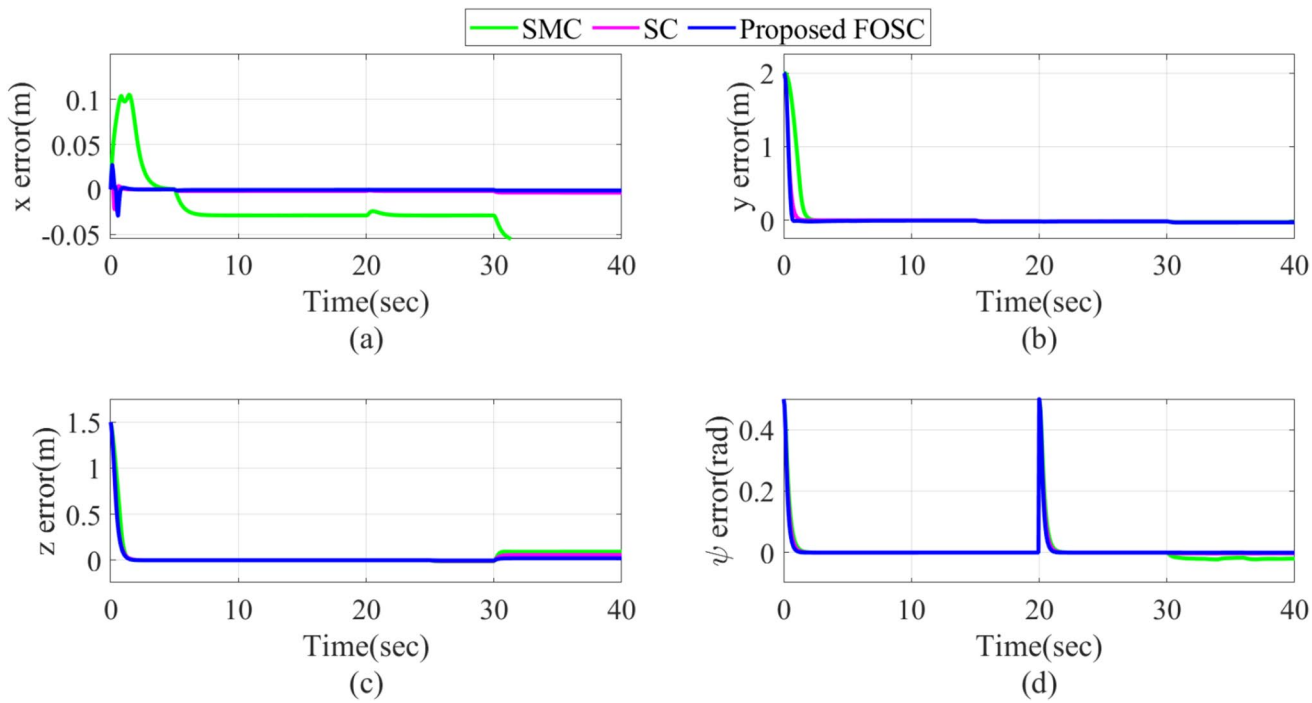


Fig. 5 Tracking errors of the translational positions (x, y, z) and yaw angle (ψ) (scenario 1)

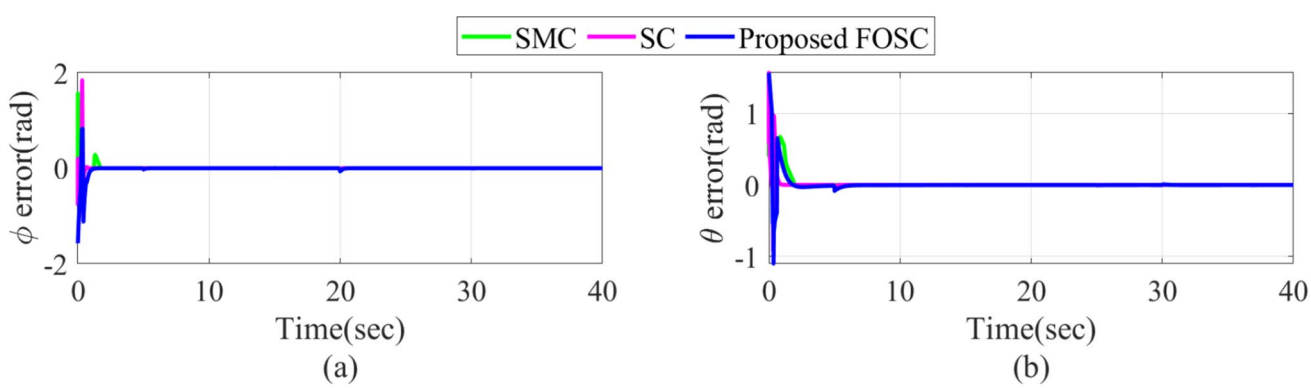


Fig. 6 Tracking errors of roll (φ) and pitch (θ) angles (scenario 1)

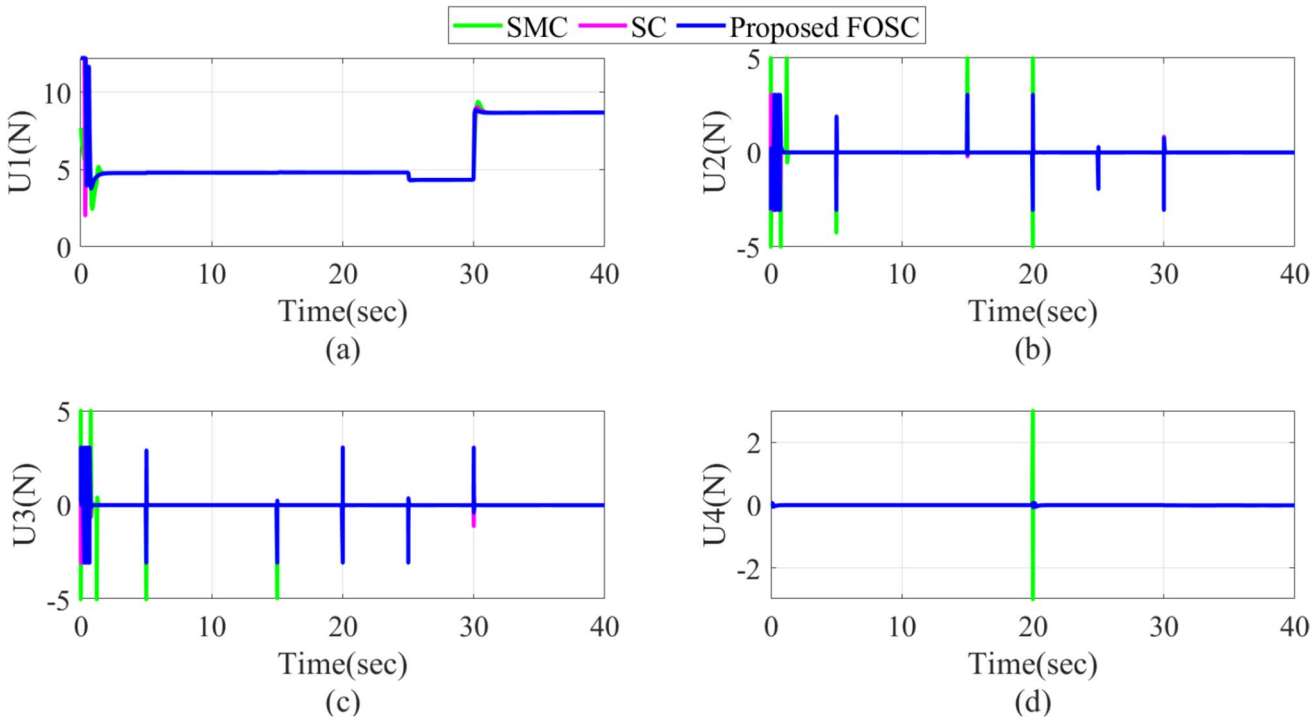


Fig. 7 Control input signals (U_1–U_4) of the quadcopter (scenario 1)

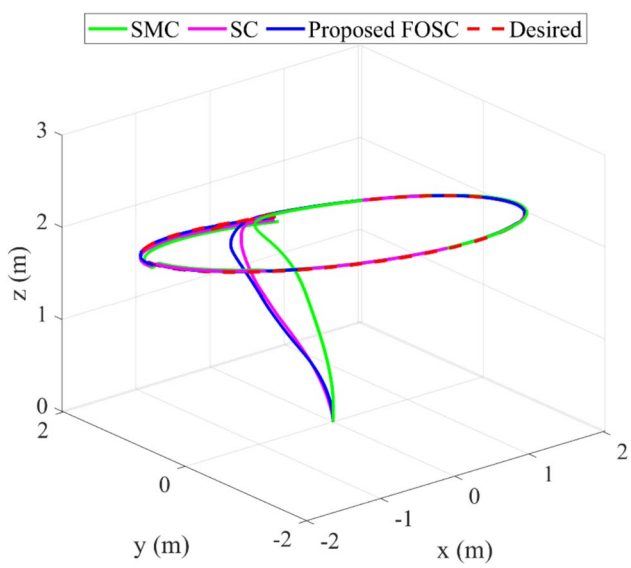


Fig. 8 The 3D trajectory tracking result of the quadcopter (scenario 1)

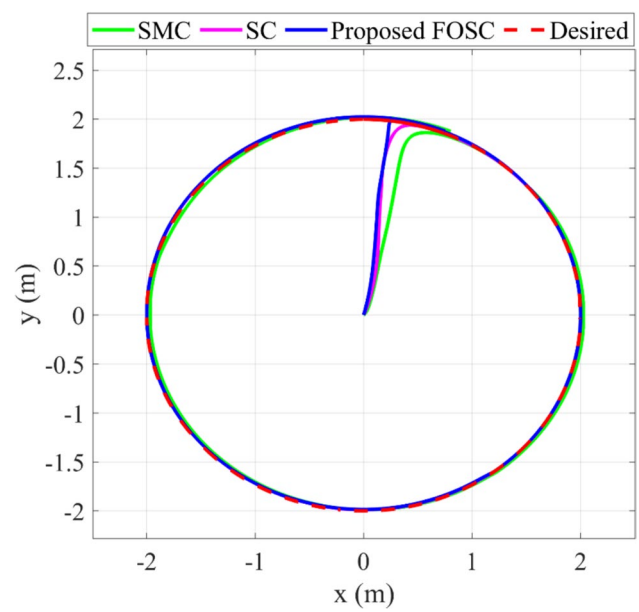


Fig. 9 The 2D trajectory tracking result of the quadcopter (scenario 1)

Table 3 RMSE performance indices (Scenario 1)

State	Proposed FOSC	SC	SMC
x	1.2635e-04	1.3769e-04	0.0021
y	0.0087	0.0088	0.0140
z	0.0063	0.0064	0.0079
φ	0.0048	0.0025	0.0018
θ	0.0063	0.0030	0.0060
ψ	0.0025	0.0026	0.0029

- If the termination conditions are not satisfied:
  - Return to Step3 to run the next iteration and remain in the optimization process.

4 Simulation results

This section presents a qualitative and quantitative comparison evaluates the efficiency of the proposed FOSC for trajectory tracking of the quadcopter UAV under external disturbances and parametric uncertainties using numerical simulations. These simulations were conducted using MATLAB software 2019b with the Simulink toolbox; the model solver used was the "ODE 1" (Euler) solver with a sampling time of Ts = 0.001 s and a total simulation time of 40 s. The computations were performed on a laptop running Windows 10 with an Intel Core i5 processor at 1.9 GHz, and 8 GB of RAM. A comparative study is presented between the proposed FOSC, SC and SMC where the sign function replaced by saturation function to reduce the chattering phenomenon [50]. The parameters of the quadcopter UAV are shown in Table 1. The parameters of the EO algorithm are given in Table 2. The optimal FOSC parameter obtained by EO are given as:

$$\Gamma_1 = [6.0867\ 0.07474.9000\ 8.77714.0372\ 3.6759],$$

$$\Gamma_2 = [0.0514\ 2.18890.0100\ 4.58641.5000\ 0.0119],$$

$$K = [0.0015\ 0.00210.0010\ 0.00900.0800\ 0.0100] \text{ and } \alpha = 0.11$$

The motor dynamics were not explicitly considered in this work. This assumption was made because the response time of brushless DC motors and electronic speed controllers (ESCs) used in quadcopters is generally much faster than the vehicle's translational and rotational dynamics. Typically, these actuators have time constants in the range of 0.01 s to 0.1 s, which makes their influence negligible under moderate flight conditions. Neglecting the motor–propeller dynamics simplifies the model and allows focusing on the main nonlinear behavior of the quadcopter and on assessing the effectiveness of the proposed fractional synergetic control law.

This modeling simplification is widely used in quadcopter control research, especially when the goal is to analyze high-level control performance rather than actuator dynamics. Many recent studies adopt the same approach, considering the motors as ideal fast-response components to reduce model complexity and simulation time [16, 20, 51]. Therefore, ignoring the motor propeller reaction time in this study does not significantly affect the controllability or validity of the obtained results.

4.1 Scenario 1

In this scenario, the quadcopter follows a circular trajectory, where the inertia moments are randomly varied within ± 20% of their nominal values, and the total mass gradually increases by 200% at 30 s, while the system is subjected to constant disturbances of d*x*, *dy*, and *dz* in m/s^2, as well as $d\varphi$, $d\theta$, and $d\psi$ in rad/s^2 expressed as:

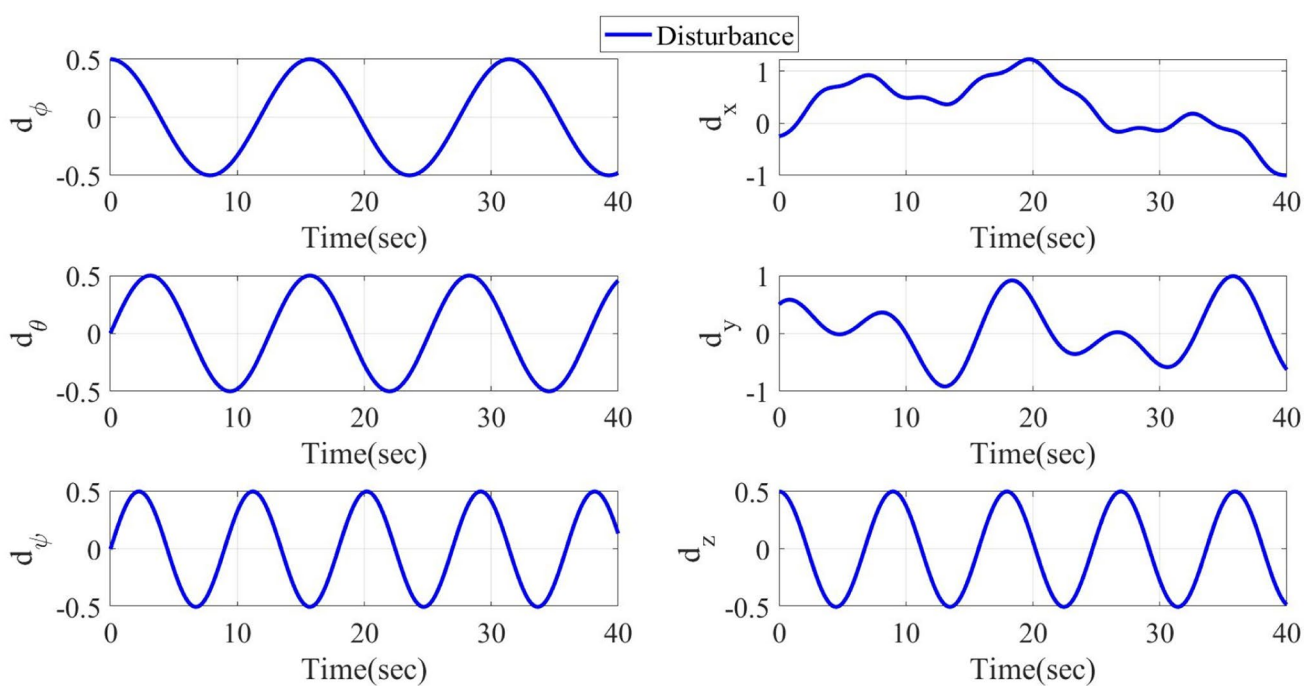


Fig. 10 Time evolution of external disturbances applied to the quadcopter (Scenario 2)

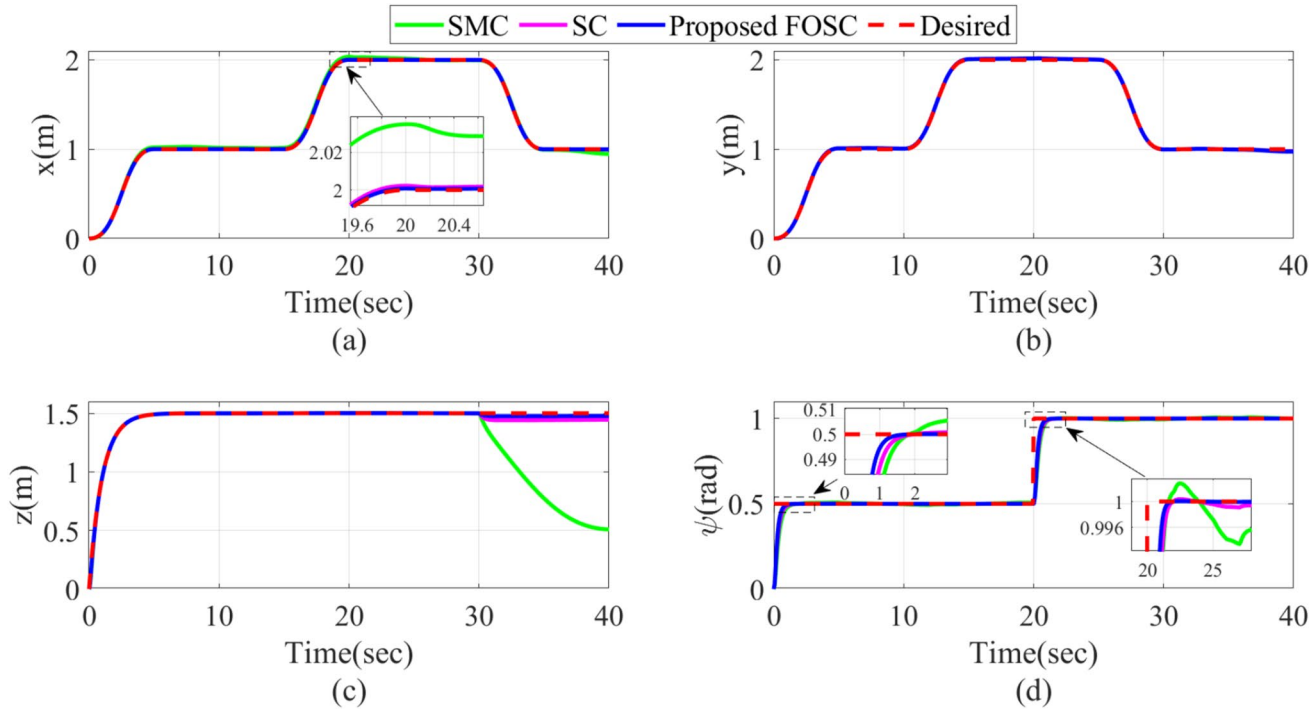


Fig. 11 Time responses of the quadcopter positions (*x, y, z*) and yaw angle (ψ) (scenario 2)

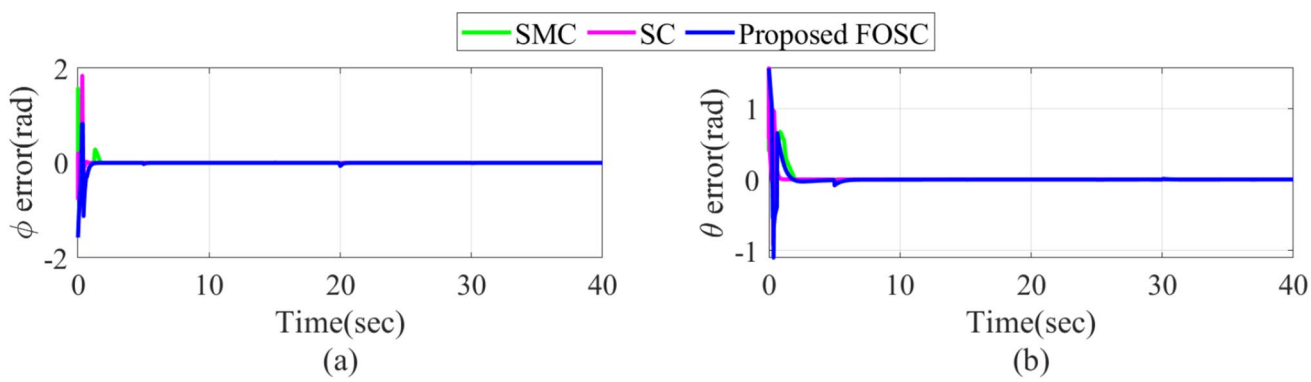


Fig. 12 Roll (φ) and pitch (θ) angular responses (scenario 2)

$$d_x = \begin{cases} 0, t \in [0,5)\text{s} \\ 1, t \in [5,40)\text{s} \end{cases}, d_y = \begin{cases} 0, t \in [0,15)\text{s} \\ 1, t \in [15,40)\text{s} \end{cases},$$
$$d_z = \begin{cases} 0, t \in [0,25)\text{s} \\ 1, t \in [25,40)\text{s} \end{cases},$$
$$d_\phi = \begin{cases} 0, t \in [0,10)\text{s} \\ 1, t \in [10,40)\text{s} \end{cases}, d_\theta = \begin{cases} 0, t \in [0,20)\text{s} \\ 1, t \in [20,40)\text{s} \end{cases},$$
$$d_\psi = \begin{cases} 0, t \in [0,30)\text{s} \\ 1, t \in [30,40)\text{s} \end{cases}. \tag{59}$$

The units for the desired trajectory of (*xd*, *yd*, *zd*) are *m*, and for ψd it is *rad*. yaw angle (ψ). Figure 3 illustrates the time responses of the quadcopter positions (x, y, z) and the yaw angle (ψ). Despite the combined parametric uncertainties, mass variation and external disturbances, the proposed FOSC maintains precise trajectory tracking with very small overshoot and fast recovery. When the mass increases at t=30 s, the transient deviation is rapidly compensated, confirming the controller's strong rejection capability. Figure 4 presents the roll and pitch angles (φ, θ), showing that the FOSC achieves smooth transitions with less oscillation compared to SMC, whereas SC exhibits slightly faster convergence in some intervals but with less stability under parameter changes.

The tracking errors in Figs. 5 and 6 further demonstrate that the FOSC minimizes steady-state deviations in position and yaw, while keeping the attitude errors bounded and free of chattering. The smoother control efforts shown in Fig. 7 emphasize the chattering-free nature of the synergetic framework enhanced by fractional calculus. This smoothness reflects a more efficient control action that avoids the

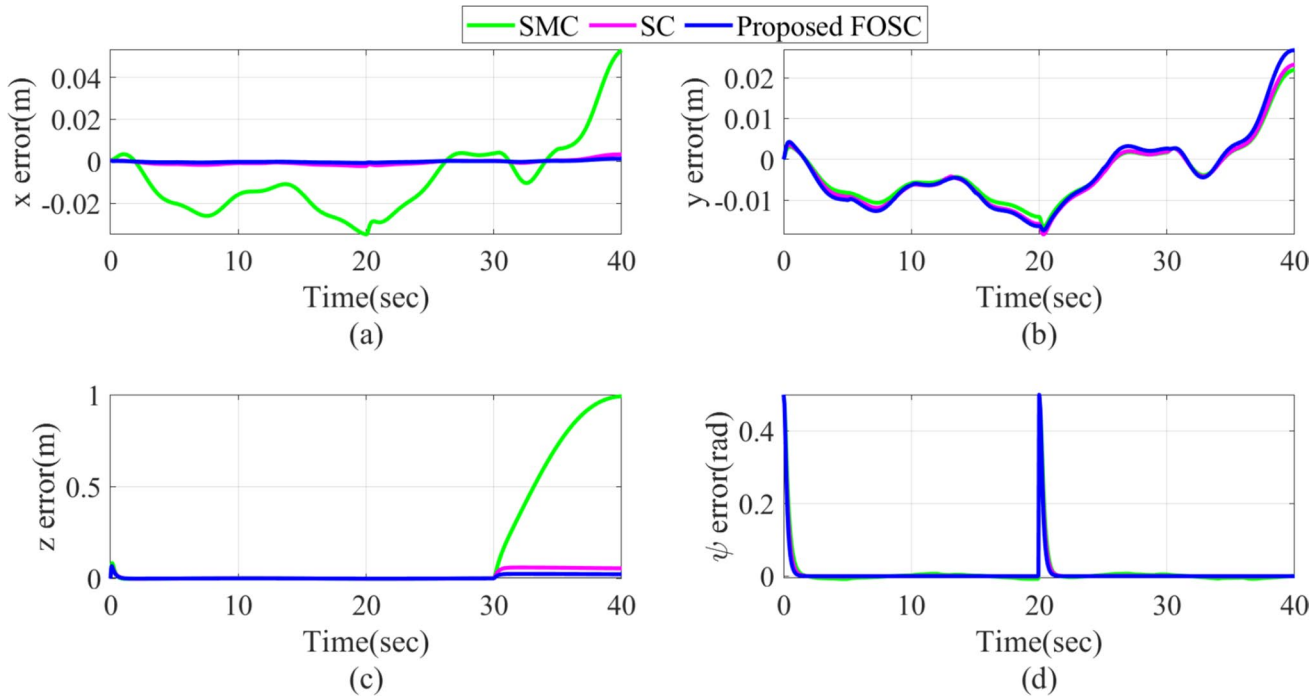


Fig. 13 Tracking errors of roll (φ) and pitch (θ) angles (scenario 2)

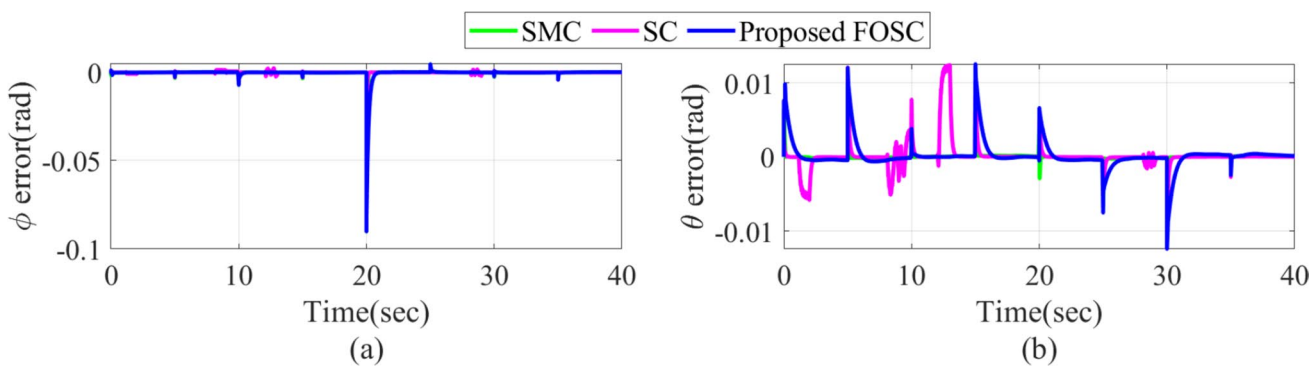


Fig. 14 Tracking errors of roll (φ) and pitch (θ) angles (scenario 2)

high-frequency switching typically observed in SMC. Figures 8 and 9 depict the 3D and 2D tracking trajectories, confirming that the FOSC keeps the UAV path tightly aligned with the desired circular trajectory, even during mass variations, while SMC exhibits noticeable drift.

The Root Mean Square Error (RMSE) values presented in Table 3 show that the proposed FOSC achieves lower tracking errors than the SC and SMC controllers in all states except for the roll (φ) and pitch (θ) angles.

4.2 Scenario 2

In this scenario, the quadcopter follows a square trajectory, where the moments of inertia are randomly changed within ± 50% of their nominal values, and the body mass increases up to 150% at t = 30 s. These external disturbances introduce dynamic variations in both translational and rotational motions, providing a meaningful way to assess the robustness and disturbance rejection capability of the proposed controller. As illustrated in Fig. 10, the time evolution of the applied disturbances reflects the typical fluctuating nature of wind-induced forces on UAVs. The mathematical representation of these disturbances is expressed as [38]:

$$\begin{cases} d_x = 0.4\sin(0.44t - 13.4) - 0.8\sin(0.1t - 3.04) + 0.08\sin(1.5t - 15\pi) \\ +0.05\sin(0.2t - 8.5),\ t \in [0, 80]\text{s}, \\ d_y = 0.5\cos(0.7t) + 0.5\sin(0.4t),\ t \in [0, 80]\text{s}, \\ d_Z = 0.5\cos(0.7t),\ \ t \in [0, 80]s, \\ d_\phi = 0.5\cos(0.4t),\ t \in [0, 80]\text{s}, \\ d_\theta = 0.5\sin(0.5t),\ t \in [0, 80]\text{s}, \\ d_\psi = 0.5\sin(0.7t),\ t \in [0, 80]\text{s}. \end{cases} \tag{60}$$

The results of the simulation are presented in Figs. 11, 12, 13, 14, 15, 16 and 17. Figure 11 shows the quadcopter

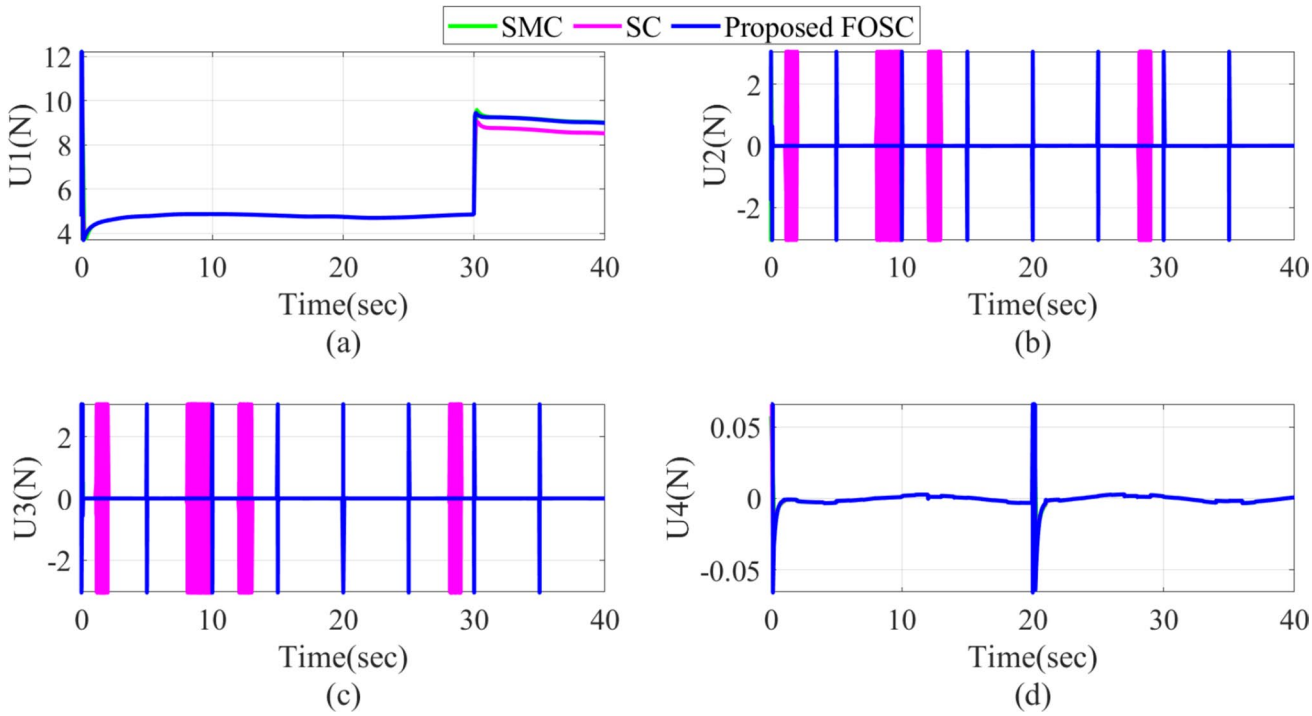


Fig. 15 Control input signals (U_1–U_4) of the quadcopter (scenario 2)

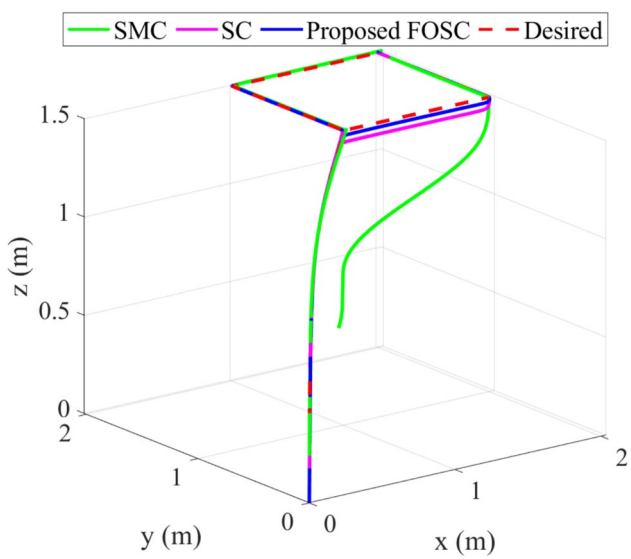


Fig. 16 The 3D trajectory tracking result of the quadcopter (scenario 2)

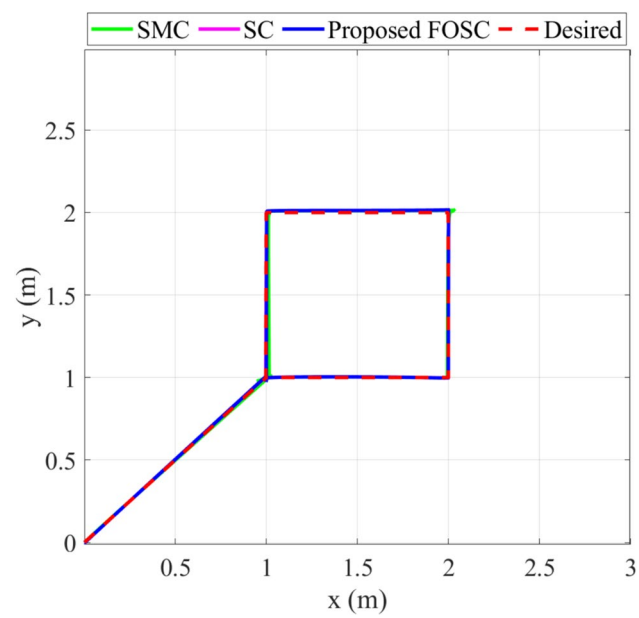


Fig. 17 2The D trajectory tracking result of the quadcopter (scenario 2)

Table 4 RMSE performance indices (Scenario 2)

State	Proposed FOSC	SC	SMC
x	2.2521e-05	6.3028e-05	9.6997e-04
y	4.8464e-04	4.5158e-04	4.1118e-04
z	6.3975e-04	0.0014	0.0179
φ	1.8647e-04	8.9950e-05	1.3458e-04
θ	8.6985e-05	1.0013e-04	2.6154e-05
ψ	0.0025	0.0025	0.0028

position (x, y, z) and the yaw angle (ψ) under time-varying disturbances, parametric uncertainties, and mass variation. It can be observed that the proposed FOSC controller maintains high tracking accuracy with very small steady-state errors and fast convergence, while SC exhibits a slight phase lag at the corners and SMC shows more significant transient deviations, especially after the mass variation at 30 s. This behavior confirms the superior robustness of the

proposed control strategy in handling both external disturbances and parameter variations. Figure 12 shows the roll and pitch quadcopter angles; the results clearly demonstrate that FOSC achieves smoother and faster convergence compared to SC and SMC. The use of fractional calculus provides additional damping and tuning flexibility, which minimizes overshoot and eliminates the high-frequency oscillations that appear in SMC.

Figures 13 and 14 illustrates the tracking errors of the main states. The results confirm that the proposed FOSC approach ensures faster error decay and smaller steady-state deviations compared to SC and SMC. The errors remain bounded even under strong disturbances and modeling uncertainties, showing the effectiveness of the proposed optimization-based design. Figure 15 show the control signals of the quadcopter. The proposed FOSC produces smooth and bounded control actions with reduced chattering, while SMC generates more abrupt control inputs that could lead to actuator wear and higher energy consumption. Around the time when the mass increases (t = 30 s), the FOSC controller quickly compensates the change and stabilizes the system faster than SC and SMC.

Figures 16 and 17 illustrate the quadcopter trajectories in 3D and 2D spaces, respectively. The results show that the proposed controller is capable of following the reference square path precisely, while SC and SMC exhibit visible deviations at the corners. Finally, Table 4 presents the RMSE values for this scenario, confirming that the proposed FOSC controller achieves the lowest tracking error in general among all compared techniques. These results demonstrate that the proposed control approach provides superior performance in terms of robustness, convergence speed, and trajectory precision, even under challenging flight conditions involving time-varying disturbances and parameter uncertainties.

The computational burden of the proposed FOSC was also analyzed. The average execution time per control cycle was found to be 1.5955×10^{-5} s, compared to 1.1209×10^{-5} s for SC and 1.1177×10^{-5} s for SMC. This minor increase is attributed to the fractional derivative computation and the number of parameters. However, the resulting computation time is still within microsecond order, which confirms that the proposed controller is suitable for real-time implementation using conventional UAV microcontrollers.

5 Conclusion

This paper has presented a robust FOSC approach for trajectory tracking of a quadcopter system under time-varying disturbances, parametric uncertainties, and mass variation. Using fractional calculus and synergistic control theory combination, the stability is verified by using the Lyapunov function. Synergistic theory is considered chattering-free, whereas fractional calculus has more freedom in the design since it has a larger stability region. Additionally, the adoption of the equilibrium optimizer allows finding the optimal controller parameters. Simulation results confirmed that the proposed controller achieves faster convergence, accurate trajectory tracking, and strong robustness when compared with sliding mode controller and synergetic controller, and the results showed smooth control signals without noticeable chattering. Although the obtained results are promising, this study is limited to simulation-based validation under the rigid-body assumption. Future work will therefore focus on experimental verification and the extension of the model to include actuator dynamics and real flight conditions.

Funding No funding was received.

Data availability Not applicable.

Declarations

Conflict of interest The authors declare no competing interests.

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